Lecture 7

Functional Dependencies

Dr. Vandana Kushwaha

Department of Computer Science

Institute of Science, BHU, Varanasi

Introduction

- A **Functional dependency** is a **constraint** between **two sets** of **attributes** from the **database.**
- Consider the whole database as being described by a single **universal relation schema *R* = {*A1*, *A2*, ... , *An*}** has ***n*** attributes ***A1*, *A2*, ...,*An*.**
- A **functional dependency**, denoted by ***X* → *Y*,** between **two sets** of attributes ***X*** and ***Y*** that are **subsets of *R*** specifies a ***constraint*** on the possible **tuples** that can form a **relation state *r*** of ***R*.**
- **Constraint** : *For all pairs of* ***tuples t1*** *and* ***t2*** *in* ***r*** *that have* ***t1[X] = t2[X],*** *they must also have* ***t1[Y] = t2[Y].***
- There is a **functional dependency from *X* to *Y*,** or that ***Y*** is **functionally dependent** on ***X*.**
- The abbreviation for **functional dependency** is **FD** or **f.d.**
- The set of attributes ***X*** is called the **left-hand side** of the **FD,** and ***Y*** is called the **right-hand side**.

Functional Dependency

- Suppose **R** is a relation, **X,Y** are attributes over **R**, **t1,t2** are tuples in the relation **R:**

	X	Y
t1	a1	b1
t2	a2	b2

- The **functional dependency X→Y** holds in **R** if and only if
  - **(t1.X=t2.X) ═▶ (t1.Y=t2.Y)**
- In above **relation R** there is **no such tuple** for which the condition **t1.X=t2.X** is **true** thus the **functional dependency X→Y** holds in **R.**

Exercise 1

- Suppose you have the following table:

A	B
a1	b1
a2	b1

  - **A→B** is **true** (B is Functionally Dependent on A) because there is **no such pair of tuples** for which **(t1.A=t2.A)** .

Exercise 2

- Suppose you have the following table:

A	B
a1	b1
a1	b1

  - **A→B** is **true** (B is Functionally Dependent on A) because there is a pair of tuples for which **(t1.A=t2.A)** ═▶ **(t1.B=t2.B)**

Exercise 3

- Suppose you have the following table:

A	B
a1	b1
a1	b2

  - **A→B** is **False** (B is Not FD on A) because there is a pair of tuples for which **(t1.A = t2.A)** but **(t1.B ≠ t2.B)**

Exercise 4

- Suppose you have the following table:

A	B	C
a_1	b_1	c_1
a_1	b_1	c_2
a_2	b_1	c_1
a_2	b_1	c_3

- List all **functional dependencies** satisfied by the relation of above Figure.
- **A→B** is **True because** there are pair of tuples for which:
  - **(t1.A=t2.A) ═► (t1.B=t2.B)** and **(t3.A=t4.A) ═► (t3.B=t4.B)**
- **C→B** is **True because** there are pair of tuples for which:
  – **(t1.C=t3.C) ═► (t1.B=t3.B)** and
  – for tuples **t2** and **t4** no such pair exist for which the values of **t2.C** and **t4.C** match.
- A dependency that logically implied by **A→B** and **C→B** : **AC → B.**

Example 5

- Let us consider the **relation *r***, to see which **functional dependencies** are satisfied.

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_2	c_1	d_2
a_2	b_2	c_2	d_2
a_2	b_2	c_2	d_3
a_3	b_3	c_2	d_4

- Observe that ***A* → *C*** is satisfied.
- There are **two tuples** that have an ***A* value** of ***a*1.**
- These tuples have the **same *C* value**—namely, ***c*1.**
- Similarly, the **two tuples** with an ***A* value** of ***a*2** have the **same *C* value, *c*2.**
- There are no other pairs of distinct tuples that have the same *A* value.
- The **functional dependency *C* → *A*** is **not satisfied**, however.
- As there is a pair of tuples ***t*1** and ***t*2** such that ***t*1[*C*] = *t*2[*C*], but *t*1[*A*] != *t*2[*A*].**

Example 6

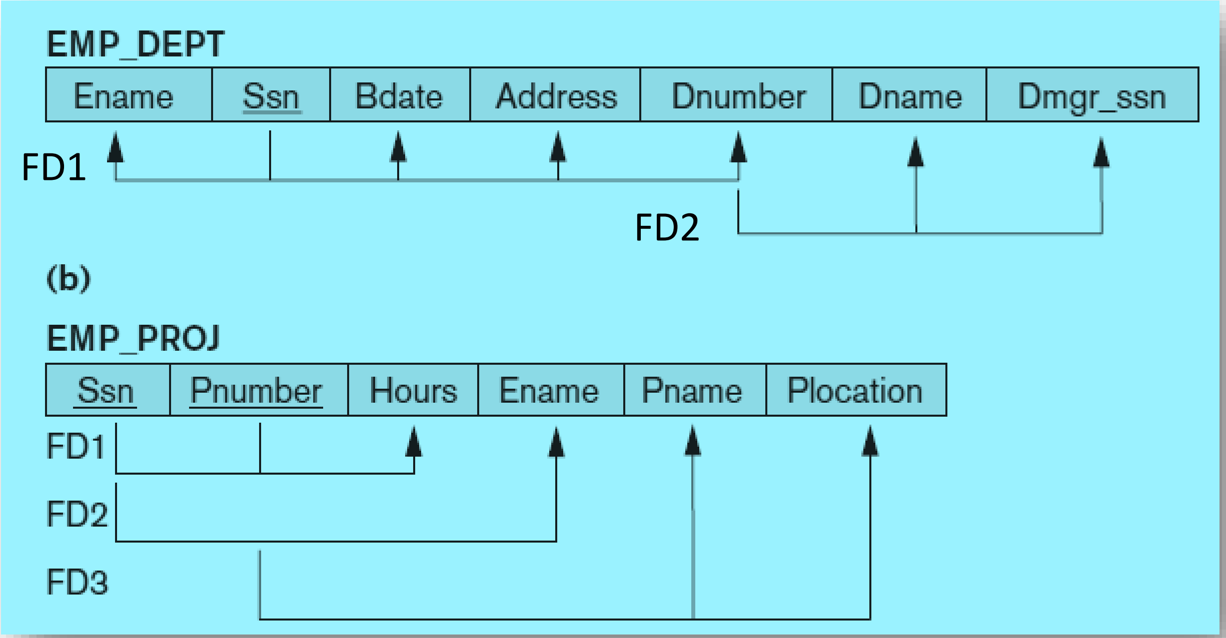
- The **functional dependency *AB* → *D*.**
- Observe that there is **no pair of distinct tuples *t1*** and ***t2*** such that ***t1*[*AB*] = *t2*[*AB*].**
- Therefore, if ***t1*[*AB*] = *t2*[*AB*],** it must be that ***t1* = *t2*** and, thus, ***t1*[*D*] = *t2*[*D*].**
- So, ***r*** satisfies ***AB* → *D*.**

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_2	c_1	d_2
a_2	b_2	c_2	d_2
a_2	b3	c_2	d_3
a_3	b_3	c_2	d_4

Functional Dependency

- Consider the relation schema **EMP_PROJ**; the following **functional dependencies** should hold:
  - **a.** Ssn → Ename
  - **b.** Pnumber → {Pname, Plocation}
  - **c.** {Ssn, Pnumber} → Hours
- These **functional dependencies** specify that:
  - **(a)** the value of an employee's Social Security number (**Ssn**) **uniquely determines** the **employee name** (Ename), alternatively, we say that **Ename** is **functionally determined by** (or functionally dependent on) **Ssn.**
  - **(b)** the value of a project's number (**Pnumber) uniquely determines** the **project name (Pname)** and **location (Plocation),** and
  - **(c)** a combination of **Ssn and Pnumber** values **uniquely determines** the **number of hours** the employee currently works on the project per week (Hours).

Diagrammatic Notation for FDs



Trivial Functional Dependency

- Some **functional dependencies** are said to be **trivial** because they are **satisfied by all relations.**
- For example, ***A* → *A*** is satisfied by **all relations** involving **attribute *A*.**
- We see that, for all tuples ***t*1** and ***t*2** such that ***t*1[*A*] = *t*2[*A*],** it is the case that ***t*1[*A*] = *t*2[*A*].**
- Similarly, ***AB* → *A*** is satisfied by **all relations** involving **attribute *A*.**
- In general, a **functional dependency** of the form $\alpha \rightarrow \beta$ is **trivial** if $\beta \subseteq \alpha$.

Inference Rules for FD

- The notation **F |=X → Y** to denote that the **functional dependency X→Y** is **inferred from** the **set of functional dependencies F.**
- The following **six rules IR1** through **IR6** are well-known **inference rules** for functional dependencies:

1. **IR1 (reflexive rule)1**: If $X \supseteq Y$, then $X \rightarrow Y$.
2. **IR2 (augmentation rule)2**: $\{X \rightarrow Y\} \models XZ \rightarrow YZ$.
3. **IR3 (transitive rule):** $\{X \rightarrow Y, Y \rightarrow Z\} \models X \rightarrow Z$.
4. **IR4 (decomposition, or projective, rule):** $\{X \rightarrow YZ\} \models X \rightarrow Y$.
5. **IR5 (union, or additive, rule):** $\{X \rightarrow Y, X \rightarrow Z\} \models X \rightarrow YZ$.
6. **IR6 (pseudotransitive rule):** $\{X \rightarrow Y, WY \rightarrow Z\} \models WX \rightarrow Z$.

- **Inference rules IR1** through **IR3** are known as **Armstrong's inference rules**.

Inferred Functional Dependency

- Let ***F denote*** the **set** of **functional dependencies** that are specified on **relation schema *R*.**
- Typically, the schema designer specifies the **functional dependencies** that are ***semantically obvious*.**
- However, numerous **other functional dependencies hold** in ***all*** legal relation instances among sets of attributes that can be **derived** from and satisfy the dependencies in ***F*.**
- Those **other dependencies** can **be *inferred* or *deduced*** from the **FDs in *F*.**
- **Example:** Dept_no → Mgr_ssn and Mgr_ssn→Mgr_phone **infer** Dept_no → Mgr_phone
- This is an **inferred FD** and need *not* be explicitly stated in addition to the two given FDs.

Closure of FD

- Formally, the **set of all dependencies** that **include *F*** as well as **all dependencies that can be inferred from *F*** is called the **closure of *F*;** it is denoted by F^+.
- **Example:** Consider the set of FDs **F:**
- F = {Ssn → {Ename, Bdate, Address, Dnumber}, Dnumber → {Dname, Dmgr_ssn} }
- Some of the **additional functional dependencies** that we can ***infer* from *F*** are the following:
  - Ssn → {Dname, Dmgr_ssn}
  - Ssn → Ssn
  - Dnumber → Dname
- Thus F^+ = **{**Ssn → {Ename, Bdate, Address, Dnumber}, Dnumber → {Dname, Dmgr_ssn}, Ssn → {Dname, Dmgr_ssn}, Ssn → Ssn, Dnumber → Dname **}**

Closure of FD

- **Example:**
- Let **R=**(A,B,C,G,H,I) is a relational schema with set of functional dependencies **F=**(A->B, A->C, CG->H, CG->I, B->H)
- Then members of **F^+** are:
  - **A->H,** since A->B & B->H hold **(transitivity rule)**
  - **CG->HI**, since CG->H & CG-> I **(union rule)**
  - **AG->I,** since A->C, CG->I (**pseudotransitivity rule)**
- Thus **F^+**={A->B, A->C, CG->H, CG->I, B->H, A->H, CG->HI, AG->I}

Closure of attribute under a FD

- **Algorithm:** Determining X^+, the **Closure of *X*** under ***F.***
- **Input:** A **set *F*** of **FDs** on a **relation schema R**, and a **set of attributes *X*,** which is a **subset of R.**

```
X⁺ := X;
repeat
    oldX⁺ := X⁺;
    for each functional dependency Y → Z in F do
        if X⁺ ⊇ Y then X⁺ := X⁺ ∪ Z;
until (X⁺ = oldX⁺);
```

Closure of Attribute under a FD

- Consider a **set *F*** of functional dependencies that should hold on **EMP_PROJ:**
- ***F*** = **{**Ssn → Ename, Pnumber → {Pname, Plocation}, {Ssn, Pnumber} → Hours**}**
  - **{Ssn}** $^+$ = Ssn
  - **{Ssn}** $^+$ = {Ssn, Ename}
  - **{Pnumber}** $^+$ = {Pnumber, Pname, Plocation}
  - **{Ssn, Pnumber}** $^+$ = {Ssn, Pnumber, Ename, Pname, Plocation, Hours}

```
X⁺ := X;
repeat
    oldX⁺ := X⁺;
    for each functional dependency Y → Z in F do
        if X⁺ ⊇ Y then X⁺ := X⁺ ∪ Z;
until (X⁺ = oldX⁺);
```

Applications of Attribute Closure

- There are **several uses** of the ***Attribute Closure algorithm:***
- **Testing of Superkey**
  - To test if $\boldsymbol{\alpha}$ is a **superkey**, we **compute** $\boldsymbol{\alpha^+}$, and **check if** $\boldsymbol{\alpha^+}$ **contains all attributes of** $\boldsymbol{R}$**.**
- **Testing the existence of a Functional Dependency**
  - We can check if a **functional dependency** $\boldsymbol{\alpha \rightarrow \beta}$ **holds** (or, in other words, is in $\boldsymbol{F^+}$), by checking if $\boldsymbol{\beta \subseteq \alpha^+}$.
  - That is, we **compute** $\boldsymbol{\alpha^+}$ by using **attribute closure**, and then **check** if it **contains** $\boldsymbol{\beta}$**.**

Example

- Consider the schema ***R* = (*A, B, C, G, H, I*)** and the **set *F*** of functional dependencies:
- ***F***=*{A → B, A → C, CG → H, CG → I, B → H}* and
- **F⁺**=(A->B, A->C, CG->H, CG->I, B->H, A->H, CG->HI, AG->I)
- Check **AG** is **super key** or not?
- **Solution:**
  - **AG^+**= AG
  - Consider the FD A->B , here A ⊆ AG^+ thus AG^+ = AG ∪ B=AGB
  - Consider the FD A->C, here A ⊆ AG^+ thus AG^+ = AGB ∪ C=AGBC
  - Consider the FD CG->H, here CG ⊆ AG^+ thus AG^+ = AGBC ∪ H= AGBCH
  - Consider the FD CG->I, here CG ⊆ AG^+ thus AG^+ = AGBCH ∪ I=AGBCHI
  - Consider the FD B->H, here B ⊆ AG^+ thus **AG^+** = AGBCHI ∪ H=**AGBCHI**(old AG^+)
  - As **AG^+** contains **all the attribute of R** thus **AG** is **a super key of R.**

Example

- Consider a relation schema **R=(A,B,C,D,E,F,G)** with set of Functional dependencies **F={A→B,BF→C,AB→F,D→E}** Which of the following **FDs** are in F^+:

  A→F, DF→E, B→E, A→G, ABC→C

Solution:

Test for *A→F*

$A^+ = AB$ *using* A→B

$\quad = ABF$ *using* AB→F

$\quad = ABCF$ *using* BF→C

As ***F ⊆ A⁺*** thus *A→F exist in* F^+.

Example

F={A→B,BF→C,AB→F,D→E}

Test for DF→*E*

DF^+= DF

= DFE using D→E

As ***E ⊆ DF⁺*** thus DF→*E exist in* F^+.

Test for B→*E*

B^+= B

As ***E ⊄ B⁺*** thus **B**→*E does not exist in* F^+.

Test for A→*G*

A^+= ABCF

As ***G ⊄ A⁺*** thus **A→*G*** *does not exist in* F^+.

Example

F={A→B,BF→C,AB→F,D→E}

Test for ABC→*C*

ABC$^+$*= ABC*

= ABCF using AB→F

As ***C*** $\subseteq$ ***ABC***$^+$ thus **ABC→*C*** *exist in* F$^+$.

Cover of Functional Dependencies

- A **set** of **functional dependencies** ***F*** is said to **Cover** another **set** of **functional dependencies** ***E*** if **every FD in** ***E*** is also **in** F^+ **i.e.** $E \subseteq F^+$
- That is, if **every dependency in** ***E*** can be **inferred from** ***F*** ; alternatively, we can say that ***E* is covered by *F*.**
- We can **determine** whether ***F* covers *E*** by calculating X^+ *with respect to* ***F*** for each **FD** $X \rightarrow Y$ ***in E***, and then checking whether this X^+ **includes** the **attributes in** ***Y*.**
- If this is the case **for *every* FD in** ***E***, then ***F* covers *E*.**

Equivalence of Sets of Functional Dependencies

- **Two sets** of **functional dependencies** ***E*** and ***F*** are **equivalent** if $E^+ = F^+$.
- Therefore, **equivalence** means that **every FD** in ***E*** can be **inferred from *F***, and **every FD in *F*** can be **inferred from *E*;**
- That is, ***E* is equivalent to *F*** if **both** the conditions**- *E* covers *F*** *and* ***F* covers *E*- hold.**
- **Example:** the following two sets of **FDs** are **equivalent:**
  - $\boldsymbol{F} = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$
  - $\boldsymbol{G} = \{A \rightarrow CD, E \rightarrow AH\}$

  As **F covers G** and **G covers F** both are **true.**

Equivalence of Sets of Functional Dependencies

- **Exercise:** Show that the following two sets of FDs are **equivalent:**
  - $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$
  - $G = \{A \rightarrow CD, E \rightarrow AH\}$.
- **Determining whether F covers G**
  - $A^+ = \{ A \ C \ D \}$ // closure of left side of A → CD w.r.t. set **F**
  - Here $CD \subseteq A^+$
  - $E^+ = \{ A \ C \ D \ E \ H \}$ // closure of left side of E → AH w.r.t. set **F**
  - Here $AH \subseteq E^+$
- **Thus F covers G.**

Equivalence of Sets of Functional Dependencies

- $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$
- $G = \{A \rightarrow CD, E \rightarrow AH\}$.

- **Determining whether G covers F**
  - $A^+ = \{ A \ C \ D \}$ // closure of left side of $A \rightarrow C$ w.r.t. set **G**
  - Here $C \subseteq A^+$
  - $(AC)^+ = \{ A \ C \ D \}$ // closure of left side of $AC \rightarrow D$ w.r.t. set **G**
  - Here $D \subseteq (AC)^+$
  - $E^+ = \{ E \ A \ C \ D \ H \}$ // closure of left side of $E \rightarrow AD$ w.r.t. set **G**
  - Here $AD \subseteq E^+$
  - $E^+ = \{ E \ A \ C \ D \ H \}$ // closure of left side of $E \rightarrow H$ w.r.t. set **G**
  - Here $H \subseteq E^+$
  - **Thus G covers F.**
  - **It means both F and G are equivalent.**

Extraneous attributes

- An **attribute** of a **functional dependency** is said to be **extraneous** if we can **remove it** without **changing the closure** of the **set** of **functional dependencies**.
- **Formal definition of extraneous attributes:**
- Consider a **set *F* of functional dependencies** and the functional dependency ***α →β*** in ***F*.**
- ***Case 1* *(*extraneous attribute A is on *LHS of α →β)*:**
  - To find if an **attribute A** in **α** is **extraneous** or not.
  - **Step 1**: Find **$(\{\alpha\} - A)^+$** using the dependencies of **F.**
  - **Step 2**: If **$(\{\alpha\} - A)^+$** contains **all the attributes** of **β**, then **A is extraneous**.

Extraneous attributes

- ***Case 2* (extraneous attribute A is on *RHS of* $\alpha \rightarrow \beta$ *)*:**
  - To find if an **attribute A** in **β** is **extraneous** or not.
  - **Step 1**: Find $\boldsymbol{\alpha^+}$ using the dependencies in **F'** where

    F' = **$(F - \{\alpha \rightarrow \beta\}) \cup \{ \alpha \rightarrow (\beta - A) \}$.**
  - **Step 2**: If $\boldsymbol{\alpha^+}$ **contains A**, then **A is extraneous**.

Example 1

- **Question:** Given a relational schema **R(P,Q,R)** and set **F** = **{**P→Q, PQ→R**}**. Is **Q** extraneous in **PQ→R?**
- ***Step 1:*** *Find **({α} – A)⁺** using the dependencies of **F.***
  - Here, **α** is **PQ**. So find **(PQ – Q)⁺,** ie., **P⁺** (closure of P).
  - P^+= PQ since P→Q
  - P^+= PQR since PQ→R
  - Hence, the closure of P is **PQR.**
- ***Step 2:*** *If **({α} – A)⁺** contains all the attributes of **β**, then **A** is **extraneous.***
  - **(PQ – Q)⁺** contains **R.**
  - Hence, **Q is extraneous in PQ→R**.

Example 2

- **Question:** Given a relational schema **R(P,Q,R)** and set **F = {P→QR, Q→R}.** Is **R** extraneous in **P→QR?**
- **Step 1:** Find $\boldsymbol{\alpha}^+$ using the dependencies in **F'** where **F'** = (F – {α → β}) U { α → (β – A) }.
- **F'** = ({P→QR, Q→R} – {P→QR}) U {P→(QR-R)} = ({Q→ R} U {P→Q}) = **{Q→R, P→Q}**
- Find **(P)**$^+$ closure of P using the **F'** .
- **P**$^+$= PQ since P→Q
- **P**$^+$= **PQR** since Q→R
- **Step 2:** If $\boldsymbol{\alpha}^+$ contains **A,** then **A** is **extraneous.**
- **Since P**$^+$ contains **R,** hence, **R** is **extraneous** in **P→QR.**

Finding Key Attribute

- **METHOD 1**
- **Algorithm :** Finding a **Key** ***K*** for ***R*** Given a set ***F*** of Functional Dependencies.
- **Input:** A **relation** ***R*** and a set of functional dependencies ***F*** on the attributes of ***R*.**

Step 1. Set ***K* := *R*.**

Step 2. For each attribute ***A*** in ***K*** compute $(K - A)^+$ with respect to ***F*;**

  - if $(K - A)^+$ contains **all the attributes** in **R**, then set **K := K – {A} };**

Note: This algorithm **determines** **only one key** out of the possible **candidate keys** for **R;** the **key** returned depends on the **order** in which **attributes** are **removed** from **R** in **step 2.**

Finding Key Attribute

- **Exercise**
- **Given a** Relation R(A,B,C,D,E) and set of Functional Dependencies **F ={** A→B , BC→D, D→E}. Find a **candidate key** using the algorithm.
- **Solution**
- Start with all attributes.
- K={A,B,C,D,E}
- **Remove A**
- Compute
- $(K-A)^{+}$={B,C,D,E}, It **does not contain A**, therefore we can not remove **A.**

Finding Key Attribute

- **Exercise**
- **Given a** Relation R(A,B,C,D,E) and set of Functional Dependencies **F ={** A→B , BC→D, D→E}. Find a **candidate key** using the algorithm.
- **Solution**
- **Remove B**
- Compute
- $(K-B)^{+}=\{A,B, C,D,E\}$
- Contains all attributes of **R**
- Thus **B is redundant**
- Update K={A,C,D,E}

Finding Key Attribute

- **Exercise**
- **Given a** Relation R(A,B,C,D,E) and set of Functional Dependencies **F ={** A→B , BC→D, D→E}. Find a **candidate key** using the algorithm.
- **Solution**
- **Remove C**
- Compute
- $(K-C)^{+}=\{A,B, D,E\}$
- It does not contain C, thus we can not remove C.

Finding Key Attribute

- **Exercise**
- **Given a** Relation R(A,B,C,D,E) and set of Functional Dependencies **F ={** A→B , BC→D, D→E}. Find a **candidate key** using the algorithm.
- **Solution**
- **Remove D**
- Compute
- $(K-D)^{+}= \{A,B,C,D,E\}$
- Contains all attributes, thus **D is redundant**
- Update K={A,C,E}

Finding Key Attribute

- Exercise
- **Given a** Relation R(A,B,C,D,E) and set of Functional Dependencies **F ={** A→B , BC→D, D→E}. Find a **candidate key** using the algorithm.
- Solution
- **Remove E**
- Compute
- $(K-E)^+= \{A,B,C,D,E\}$
- Contains all attributes, thus **E is redundant**
- Update K={A,C}
- Thus **Candidate key** for relation **R** is **AC**.

Finding Key Attribute

- **METHOD 2**
- Determine all **essential attributes** of the **given relation**.
- **Essential attributes** are those **attributes** which are **not present on RHS** of **any functional dependency.**
- **Essential attributes** are always a **part of** every **Candidate key.**
- Now, following **two cases** are possible-
- **Case-1**
- If **all essential attributes together** can **determine all** remaining **non-essential attributes,** then-
  - The **combination of essential attributes** is the **candidate key.**
  - And it is the **only possible candidate key.**

Finding Key Attribute

- **Case-2**
- If **all essential attributes together can not determine** all remaining **non-essential attributes**, then-
  - The **set** of **essential attributes** and **some non-essential attributes** will be the **candidate key(s).**
  - In this case, **multiple candidate keys** are possible.
  - To find the **candidate keys,** we check **different combinations** of **essential and non-essential attributes.**

- ***Number of possible Super keys =** $2^{(N - (\text{size of candidate key}))}$
- Where **N** is the **number of attributes** in relation **R.**

Example 1

- **Ex1.** Let **R** = (A, B, C, D, E, F) be a relation scheme with the following dependencies- **F**={ C → F, E → A, EC → D, A → B } Find the **candidate key(s).** How many possible **super keys** are there?
- **Solution:**
- **Essential attributes** of the relation are- C and E.
- So, attributes **C and E** will definitely be a part of **every candidate key.**
- **{ CE }$^+$**

  = { C , E }

  = { C , E , F } (Using C → F)

  = { A , C , E , F } (Using E → A)

  = { A , C , D , E , F } (Using EC → D)

  = { A , B , C , D , E , F } (Using A → B)

Example 1

- Thus **CE** is the **only possible** **candidate key** of the relation **R** = (A, B, C, D, E, F)
- As **number of possible** **super keys** is $\mathbf{2^{(N - (\text{size of candidate key}))}}$
- Here **N=6**, size of candidate key=**2**
- Thus **possible** **Super keys** is $2^{(6-2)}$= **16 super keys.**

Example 2

Ex2. Let **R** = (A, B, C, D, E) be a relation scheme with the set of functional dependencies **F**={AB → C, C → D, B → E}, Determine the total number of **candidate keys** and **super keys.**

Solution:

- **Essential attributes** of the relation are- **A and B.**
- So, attributes A and B will definitely be a part of every **candidate key.**
- **$\{ AB \}^+$**

  = { A,B,C,D,E }

  Thus **AB** is the **only possible candidate key** of the relation.

  Possible super keys is $2^{(5-2)}$= **8 super keys.**

Example 3

Ex3. Consider the relation scheme **R**(E, F, G, H, I, J, K, L, M, N) and the set of functional dependencies **F**= EF→ G, F→ I J, EH → KL , K → M , L → N What is the **key** for **R**?

Solution:

- **Essential attributes** of the relation are- **E, F** and **H.**
- So, attributes **E, F** and **H** will definitely be a part of every **candidate key.**
- **{ EFH }$^+$** = { EFGHIJKLMN }
- So, **EFH** is the **only possible** **candidate key** of the **relation R**.
- **Possible super keys** is **2**$^{(10-3)}$= **128 super keys.**

Example 4

Ex4. Consider the relation scheme **R**(A, B, C, D, E, H) and the set of functional dependencies **F**= A → B, BC → D, E → C, D → A . What are the **candidate keys** of **R**?

Solution:

- **Essential attributes** of the **relation R** are- **E** and **H.**
- So, attributes **E** and **H** will definitely be a part of every **candidate key.**
- **{ EH }$^+$** = { EHC } thus **EH alone is not the candidate key.**
- **{ AEH }$^+$** = {ABCDEH} , thus **AEH** is a **candidate key.**
- **{ BEH }$^+$** = {ABCDEH} , thus **BEH** is a **candidate key.**
- **{ CEH }$^+$** = {CEH} , thus **CEH** is **not a candidate key.**
- **{ DEH }$^+$** = {ABCDEH} , thus **DEH** is a **candidate key.**

Number of Possible Super keys

- **CASE 1: Only one Candidate Key**
- Number of possible Super keys =$\mathbf{2^{(N - (\text{size of candidate key}))}}$

- **CASE 2: Two Candidate Keys**
- Let a Relation **R** have attributes {a1, a2, a3,…, an} and the **candidate keys** are "**C1**" of **size x** and "**C2**" of **size y** then the possible number of **super keys**:
- = (super keys possible with candidate key C1) + (super keys possible with candidate key C2) – (super key from both C1 and C2)
- **= $2^{(n-x)} + 2^{(n-y)} - 2^{(n-z)}$**
- z= size of **super key** from both C1 and C2

Example 5

- Let a **Relation R** have attributes **{a1, a2, a3,…, an}** and the **candidate keys** are "**a1**", "**a2 a3**" then the possible number of **super keys**?

- **Number of possible super keys**
- = (super keys possible with candidate key a1) + (super keys possible with candidate key (a2 a3)) – (super keys from both a1 and (a2 a3))
- $=\mathbf{2^{(n-1)} + 2^{(n-2)} - 2^{(n-3)}}$

Example 6

- Let a **Relation R** have attributes **{a1, a2, a3,…, an}** and the **candidate keys** are "**a1 a2**", "**a3 a4**" then the possible number of **super keys?**

- **Number of possible super keys**
- = (super keys possible with candidate key (a1 a2)) + (super keys possible with candidate key (a3 a4)) – (super keys from both (a1 a2) and (a3 a4))
- $=\mathbf{2^{(n-2)} + 2^{(n-2)} - 2^{(n-4)}}$

Example 7

- Let a **Relation R** have attributes **{a1, a2, a3,…, an}** and the **candidate keys** are **“a1 a2**”, “**a1 a3”** then the possible number of **super keys?**

- **Number of possible super keys**
- = (super keys possible with candidate key (a1 a2)) + (super keys possible with candidate key (a1 a3)) – (super keys from both (a1 a2) and (a1 a3))
- $=\mathbf{2^{(n-2)} + 2^{(n-2)} - 2^{(n-3)}}$

- * Super keys from both (a1 a2) and (a1 a3) will be (a1 a2 a3).

Minimal Sets of Functional Dependencies

- A **Minimal cover** of a set of **functional dependencies F** is a set of functional dependencies ***Fc*** that satisfies the **properties:**
  - F $\subseteq$ **Fc^+,** In addition, **this property is lost** if **any dependency** from the **set *Fc* is removed.**
  - ***Fc*** must have **no redundancies** in it.
  - The **dependencies in *Fc*** are in a **standard form**(that is, they have **only one attribute** on the **right-hand side**).
- **Minimal cover** is also known as **Canonical cover.**

Formal definition of Minimal cover

- A **Minimal/Canonical cover** ***Fc*** for ***F*** is a **set of dependencies** such that:
  - ***F*** **logically implies** all **dependencies** in ***Fc*****.**
  - ***Fc*** **logically implies** all **dependencies** in ***F*****.**
  - No **functional dependency** in ***Fc*** contains an **extraneous attribute**.
  - Each **right side** of a **functional dependency** in ***Fc*** must have a **single attribute.**
  - Each **left side** of a **functional dependency** in ***Fc*** is **unique** **i.e.** there are **no two dependencies** $\alpha 1 \rightarrow \beta 1$ and $\alpha 2 \rightarrow \beta 2$ in ***Fc*** such that $\alpha 1 = \alpha 2$**.**

Significance of Canonical Cover

- Suppose that we have a **set** of **functional dependencies** ***F*** on a **relation schema** ***R.***
- Whenever a user performs an **update** on the **relation,** the **database system** must **ensure** that the **update does not violate** any **functional dependencies**.
- That is, **all the functional dependencies in *F*** are **satisfied** in the **new database state.**
- The system must **roll back** the **update** if it **violates any functional dependencies** in the **set *F*.**
- We can **reduce the effort spent** in **checking for violations by testing** a **simplified set** of **functional dependencies** that has the **same closure** as the given set.
- In other words we want a **minimal version** of **F,** that still represents **all constraints** imposed by **F.**

Algorithm to compute Canonical Cover

- ***Fc = F***
- **repeat**
  - Use the **union rule** to **replace** any dependencies in ***Fc*** of the form:
    - $\alpha_1 \rightarrow \beta_1$ and $\alpha_1 \rightarrow \beta_2$ with $\alpha_1 \rightarrow \beta_1 \beta_2$.
  - Find a **functional dependency** $\alpha \rightarrow \beta$ in ***Fc*** with an **extraneous attribute** either in α or in β.
  - /* Note: the test for extraneous attributes is done **using *Fc***, not *F* */
  - If an **extraneous attribute** is found, **delete** it from $\alpha \rightarrow \beta$.
- **until** ***Fc*** does not change.

Note: A set of **functional dependencies** can have **multiple canonical covers**!

Example

Question: Consider the following set ***F*** of functional dependencies on schema **R(*A,B,C*), F={*A* →*BC, B* → *C, A* →*B, AB* → *C*}**, Compute the **canonical cover *F^c^*.**

Solution:

- ***F^c^*** = {*A* →*BC*, *B* → *C*, *A* →*B*, *AB* → *C*}
- Applying union rule on *A* →*BC and A* →*B we get A* →*BC thus*
- *F^c^*= {*A* →*BC*, *B* → *C*, *AB* → *C*}
- **Is B extraneous in *AB* → *C*?**
- $\boldsymbol{A^+}$= *ABC thus* ***B is* extraneous** in *AB* → *C*.
- Thus after removing **B**, *AB* → *C becomes* ***A*→ *C*.**
- Thus ***F^c^***= {*A* →*BC*, *B* → *C*, *A* → *C*}

Example

- **Is B extraneous in *A →BC*?** (*F^c*= *{A →BC, B → C, A → C}*)
- We have to find **F'** = (F – {α → β}) U { α → (β – A)}
- F' = *B → C, A → C*
- ***A^+*** *= AC does not contains B thus* ***B is not*** **extraneous**.
- **Is C extraneous in *A →BC*?**
- We have to find F' = (F – {α → β}) U { α → (β – A)}
- **F'** = *B → C, A → B, A → C*
- ***A^+*** *= ABC contains C thus* ***C is*** **extraneous,** remove **C** from ***A →BC*** *which becomes* ***A→B***
- Thus ***F^c***= *{A →B, B → C, A → C} after removing the* ***redundant FD*** *A → C.*
- *Thus* ***F^c= {A →B, B → C}*** *is the* ***minimal cover.***
- ***Note:*** A **canonical cover** might **not be unique.**

Exercise

- **Q.** Consider the set of functional dependencies ***F*** = $\{A \rightarrow BC, B \rightarrow AC, \text{ and } C \rightarrow AB\}$ find all possible **Canonical covers.**